

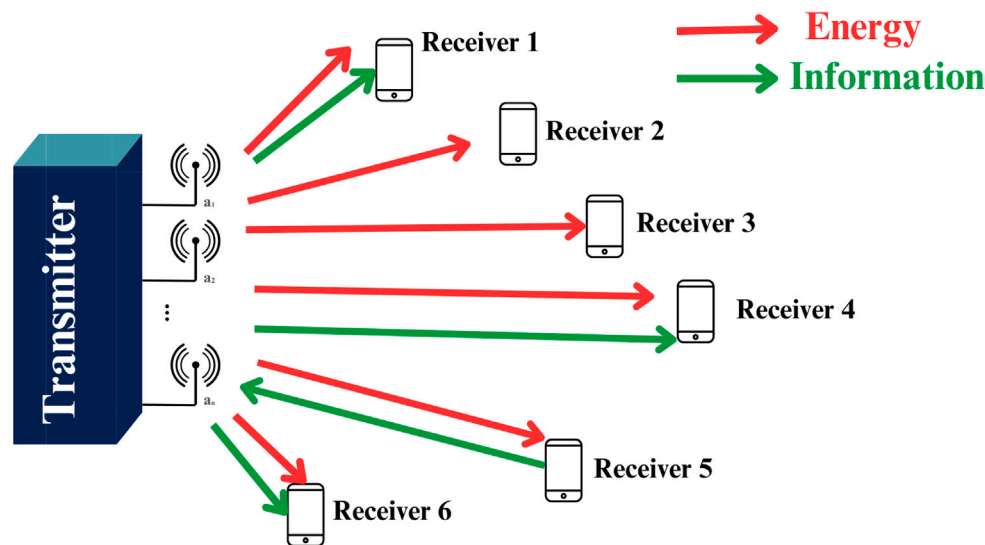


Fig. 8. Model for transfer of wireless information and energy.

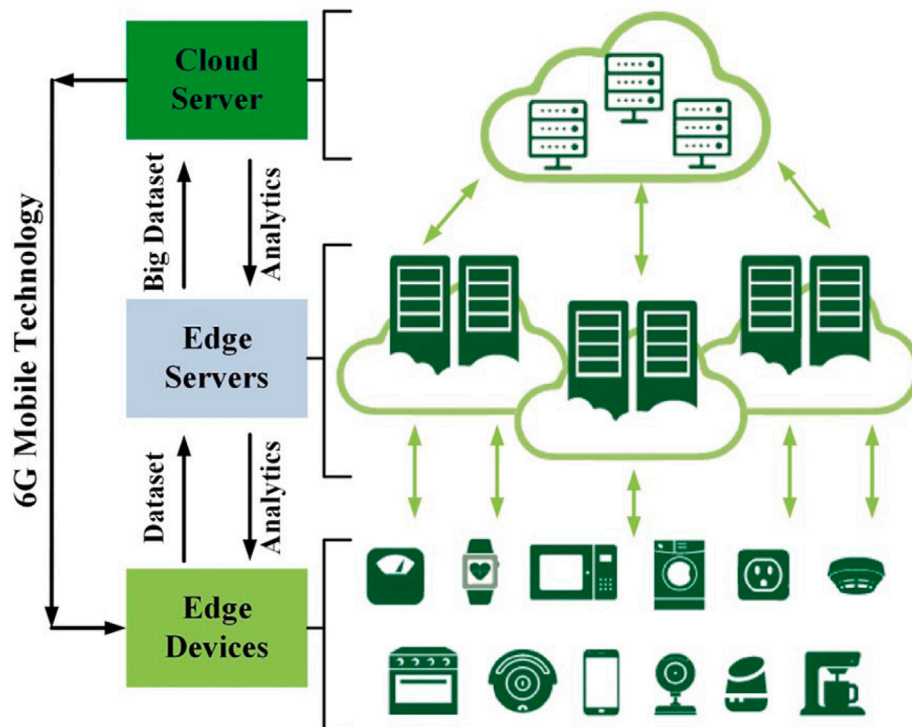


Fig. 9. Edge computing [51].

4.13. Digital twin networks

Digital Twin Networks (DTNs) are emerging as a transformative technology for 6G, enabling real-time mirroring, prediction, and optimization of physical network environments through virtual replicas. A digital twin integrates heterogeneous data sources, including sensors, network states, user mobility patterns, and environmental parameters, to maintain an accurate, continuously updated model of the network. This virtual environment can be used to test configurations, simulate failures, optimize resource allocation, and design proactive control strategies without affecting the live system. In 6G, DTNs will be essential for managing the increasing complexity introduced by

ultra-massive MIMO, 3D integrated networks (ground–air–space–sea), and AI-native communication layers. By offering predictive analytics and closed-loop control, DTNs enable self-adaptation, fault prediction, proactive load balancing, and end-to-end Quality of Service (QoS) optimization. Integrating DTNs with edge intelligence further ensures low-latency synchronization between the physical and digital domains [52]. Recent works highlight the relevance of DTNs in future networks. For instance, emerging frameworks propose DTN-enabled RAN optimization, virtual testing for UAV-assisted communication, and dynamic resource allocation for non-terrestrial networks. These studies confirm that DTNs will serve as a cornerstone in realizing fully autonomous, resilient, and intelligent 6G infrastructures [53].

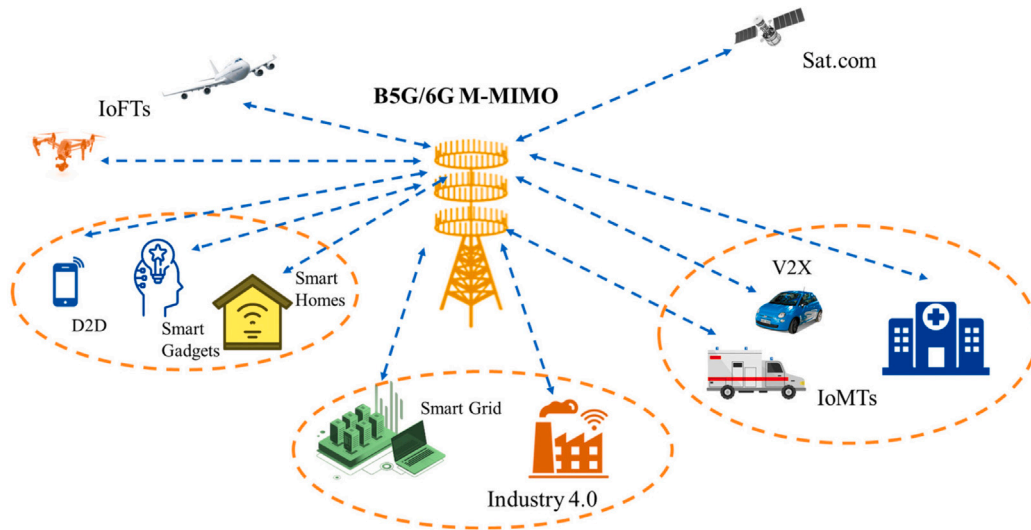


Fig. 10. Multi-antenna array-based massive MIMO for B5G/6G [55].

4.14. Free duplexing and spectrum sharing (FDSS)

Duplexing in 6G enables simultaneous data transmission and reception, overcoming inefficiencies of earlier techniques like FDD and TDD used in 1G–4G. With a full duplex system, 6G allows users to share all resources (frequency, time, and space) efficiently, achieving low latency and high throughput. Advanced spectrum sharing, supported by AI and blockchain, will enable dynamic resource management. Non-orthogonal multiple access (NOMA) further enhances spectrum efficiency, latency, and user capacity, making it a key technology for 6G communication systems [54].

4.15. Ultra massive MIMO

Ultra Massive MIMO, with its extensive antenna arrays, significantly enhances spectral efficiency and signal quality through spatial multiplexing and beamforming. In 6G, it will address capacity issues by enabling precise, high-directionality beams and mitigating interference, while also supporting high-frequency bands like THz. With over 10,000 antenna elements, 6G will achieve unprecedented spectrum efficiency and reduced propagation loss, ensuring high data rates and extended signal coverage [55]. Fig. 10 presents the Multi-Antenna Array-Based Massive MIMO for B5G/6G.

4.16. Blockchain technology

Blockchain technology provides decentralized, secure frameworks essential for managing 6G networks, addressing spectrum-sharing challenges, and enabling efficient resource utilization. Integrated with AI, it enhances QoS through smart resource allocation, scalable networks, and advanced caching. Blockchain's trust, transparency, and low-latency capabilities support secure data sharing, identity management, and transaction validation in multi-stakeholder ecosystems, making it a transformative tool for the telecommunications industry [56].

4.17. Tactile internet

The Tactile Internet enables real-time human-to-machine (H2M) interactions through haptic sensors, allowing users to remotely sense and control devices. With applications in fields like healthcare, virtual reality, and military operations, it relies on ultra-low latency and high reliability, which 6G technology is poised to deliver [57].

4.18. Quantum communication and quantum ML

Quantum computing (QC) enables parallel processing of large, multidimensional data, significantly reducing computation times for complex algorithms like successive interference cancellation (SIC). QC will enhance 6G capabilities by providing high computational power, advanced link capacity, and ultra-secure communication [58].

5. The new architecture for the core network

6G networks aim to provide energy-efficient and globally seamless wireless connectivity while addressing the limitations of current network architectures, which are inadequate for meeting the stringent requirements of future applications, such as ultra-high throughput, ultra-low latency, and enhanced reliability. Consequently, forward-looking research into next-generation network frameworks is essential. The FG NET-2030 initiative established Sub-Group 3 to design the architecture for Network 2030 [8]. However, accurately defining the future network architecture remains challenging. As a compromise, Sub-Group 3 has opted to interpret the architecture from multiple dimensions rather than proposing a unified framework. This section explores the architectural changes anticipated for 6G, as depicted in Fig. 11.

In addition to these early perspectives, recent advances highlight that 6G architecture will be fundamentally AI-native, modular, and fully integrated across terrestrial, aerial, space, and underwater domains. The architecture will rely on programmable, software-defined components and distributed intelligence to enable real-time optimization, predictive control, and autonomous operation. Emerging paradigms such as Digital Twin Networks (DTN) provide a virtual replica of the network that continuously synchronizes with physical elements, enabling proactive maintenance, fault prediction, and dynamic resource management. These digital replicas also support scenario testing and network optimization without disturbing live operations, significantly enhancing reliability and adaptability.

5.1. From terrestrial to ubiquitous 3D coverage

A key goal of next-generation network architecture is to extend communication coverage both horizontally and vertically. The current terrestrial cellular infrastructure has two major limitations: an inability to support high-altitude and deep-sea communication scenarios, which are critical for future services, and prohibitively high costs

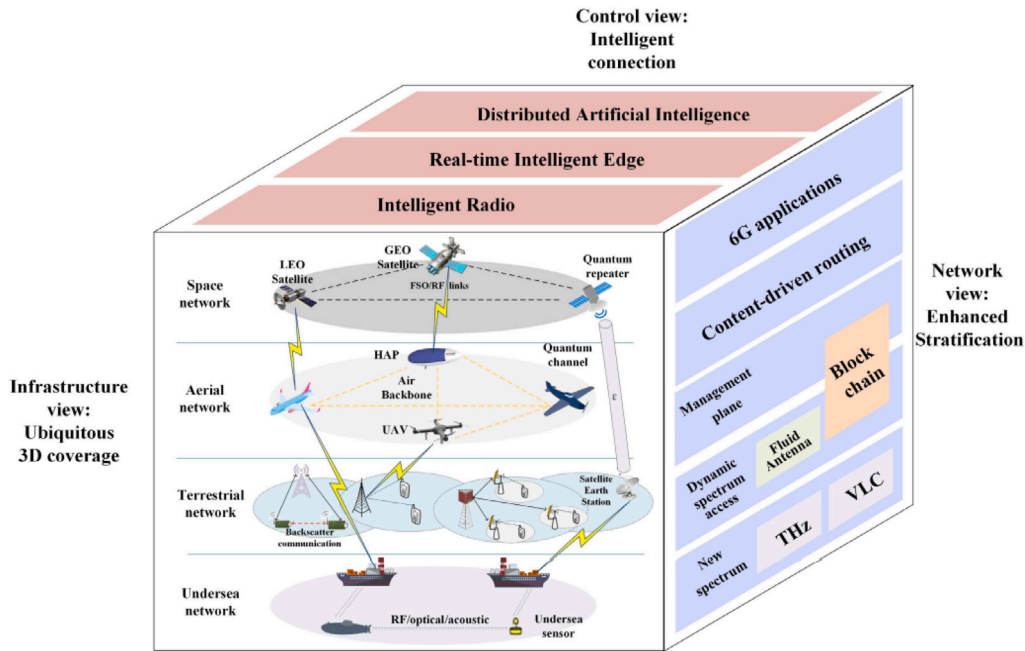


Fig. 11. Different dimensions of the architecture of 6G network [8].

associated with dense cellular networks required for global-scale connectivity. To address these limitations, 6G will integrate non-terrestrial networks to deliver comprehensive wireless coverage. So the initial concepts for Space–Air–Ground–Sea integrated communication have been discussed [59].

Beyond simple multi-layer integration, the 3D architecture of 6G must provide seamless handover across different platforms with varying mobility patterns, propagation characteristics, and link budgets. This requires new routing protocols, AI-assisted beamforming, cross-domain synchronization, and interoperability frameworks to maintain stable connectivity between satellites, HAPs, UAVs, terrestrial cells, and underwater systems.

5.1.1. Space network: LEO satellite systems

High-Throughput Satellite (HTS) systems now offer broadband Internet services with cost and bandwidth performance comparable to terrestrial networks. However, most communication satellites operate in geostationary orbit (GEO) at 35,786 km altitude, resulting in high latency and integration challenges with terrestrial mobile networks. To overcome these limitations, Non-Geostationary Orbit (NGSO) satellite systems — such as those in Low Earth Orbit (LEO) and Medium Earth Orbit (MEO) — have been proposed to provide global, low-latency, high-throughput Internet connectivity [60]. Major NGSO initiatives include: Starlink, with over 11,000 planned satellites by 2027 [8]; OneWeb, targeting a 720-satellite LEO constellation; and China's Hongyan project, expected to deploy 320 satellites by 2025.

Though large-scale deployment and integration of NGSO systems with terrestrial networks are still in progress, theoretical and simulation results demonstrate their potential. Notably, LEO satellite networks with laser and RF co-routing can outperform terrestrial fiber links in latency over distances greater than 3000 km.

To enable global connectivity, the Space-Terrestrial Integrated Network (STIN) architecture has been introduced, comprising space-based backbone networks (SBN), terrestrial networks (TN), and space-based access networks (SAN) [61]. In this model, GEO satellites (SBN) ensure wide-area coverage with inter-satellite links (ISLs), while LEO and MEO satellites (SAN) facilitate integration with HTS and terrestrial networks for ubiquitous access.

Emerging technologies play a key role in enhancing STIN functionality. Software-Defined Networking (SDN) and Information-Centric Networking (ICN) offer adaptive control, efficient configuration, and reduced latency. Additionally, Multipath TCP (MPTCP) ensures high throughput and robust mobility management through seamless handover capabilities, as validated in recent evaluations.

To ensure stable global connectivity, future LEO networks will be deeply integrated with terrestrial cores using SDN/NFV paradigms. AI-driven route prediction will mitigate Doppler effects and optimize inter-satellite link utilization, while Digital Twin models will help operators evaluate constellation performance, simulated failures, and congestion management strategies.

5.1.2. Aerial network: Flexible relay services

Aerial networks are generally divided into two primary categories: High-Altitude Platforms (HAPs), which operate in the stratosphere, and Low-Altitude Platforms (LAPs), which typically function at altitudes of a few kilometers [59].

- **High-Altitude Platforms (HAPs):** These systems provide broad coverage areas and long endurance, offering capabilities comparable to Low Earth Orbit (LEO) satellite networks.
- **Low-Altitude Platforms (LAPs):** This category includes Unmanned Aerial Vehicles (UAVs), which are characterized by rapid deployment, high maneuverability, and efficient short-range communication performance.

UAV-based LAPs can serve as relay nodes to support long-distance communication and facilitate seamless integration between terrestrial and non-terrestrial networks. Their flexibility and adaptability make them especially useful in emergency and disaster recovery situations where conventional communication infrastructure is either unavailable or severely damaged [60].

Furthermore, UAVs and HAPs may be equipped with reconfigurable intelligent surfaces (RIS) or lightweight THz transceivers, enabling aerial beam steering, temporary backhaul creation, and adaptive coverage extension. Their integration introduces new challenges such as flight-based mobility management, energy constraints, and airspace regulation, which must be addressed for safe and predictable operation.

Table 3

Comparison of underwater wireless communication technologies.

Parameter	RF	Acoustic	Optical
Attenuation	High	Lowest	Depends on turbidity
Achievable data rate	~Mbps	~Kbps	~Gbps
Latency	Moderate	High	Low
Transmission distance	<10 m	<100 km	<100 m
Power consumption	Moderate	High	Low

5.1.3. Undersea network

The potential inclusion of undersea networks in future 6G systems remains a topic of debate. Underwater wireless communication primarily utilizes three technologies: radio frequency (RF), acoustic, and optical communication [62]. A comparative analysis of these technologies is summarized in Table 3.

The deployment of undersea communication systems presents several significant challenges due to the highly unpredictable and dynamic nature of the underwater environment. Establishing stable network connectivity is inherently difficult, as water conditions can change rapidly with variations in temperature, salinity, turbidity, and ocean currents, all of which affect signal propagation and system performance. Moreover, underwater channels suffer from severe signal attenuation, particularly for radio frequency and optical communication, while acoustic signals, although capable of long-range transmission, experience low data rates and high latency. The extreme physical conditions of deep-sea environments, including high pressure, corrosion, and biofouling, further increase the risk of equipment degradation or failure. These factors make network setup, maintenance, and long-term operation considerably more complex than in terrestrial or aerial communication scenarios, highlighting the need for robust designs, adaptive communication techniques, and resilient hardware capable of withstanding harsh underwater conditions.

6. Use cases and emerging services

6G mobile technology introduces significant advancements, including high bandwidth, ultra-reliable communication, increased data rates, low latency, and an ideal environment for AI integration. These enhancements open up new application areas, ranging from military uses to meeting personal needs, particularly in scenarios requiring remote services, as highlighted during the pandemic. The demand for reliable communication with high data rates has become increasingly critical. Now we explore the opportunities and benefits of 6G technology, highlighting its practical applications. Some of these applications are presented in Fig. 12.

6.1. Connected Robotics and Autonomous Systems (CRAS)

The advancement of Collaborative Robotics and Autonomous Systems (CRAS) is a major driver in the development of 6G networks. CRAS includes emerging technologies such as autonomous vehicles, drone swarms, flying delivery systems, and robotic platforms, all of which impose stringent requirements on network performance beyond those addressed by traditional IoT services in 5G (example in Fig. 13) [63].

These applications demand ultra-low latency for real-time coordination and high throughput to handle the vast volumes of data generated by sensors, cameras, and control systems. For example, autonomous vehicles rely on seamless data exchange to enable synchronized navigation and ensure safety. Similarly, drone swarms used in defense, disaster response, or precision agriculture require rapid communication to operate collaboratively. Flying delivery systems also depend on stable, low-latency connectivity to ensure timely and efficient logistics operations [64]. By meeting these performance requirements, 6G will be a foundational enabler for next-generation CRAS deployments across a wide range of industrial and societal applications.

6.2. Extended reality

Extended Reality (XR), which encompasses Augmented Reality (AR), Mixed Reality (MR), and Virtual Reality (VR), represents a major advancement in immersive human-machine interaction. XR technologies merge physical and digital environments, enabling users to interact with computer-generated content in real time. Among them, Mixed Reality (MR) offers the most seamless integration, allowing dynamic interaction between real and virtual objects by eliminating the barriers that traditionally separate the two domains [65].

The realization of high-quality XR experiences requires ultra-low latency, high bandwidth, and reliable connectivity, capabilities that 6G is poised to deliver. Through its support for real-time, high-resolution content delivery and interactive holographic applications, 6G will unlock the full potential of XR technologies. Furthermore, the integration of edge computing and AI-driven network intelligence in 6G will ensure enhanced responsiveness, stability, and personalization of XR environments, making them suitable for applications across education, healthcare, entertainment, and industry.

6.3. Internet of Everything (IOE)

The Internet of Everything (IoE) extends the concept of the Internet of Things (IoT) by integrating people, data, processes, and devices into a unified intelligent network. Through interconnected sensing devices, IoE facilitates real-time identification, monitoring, and decision-making across diverse environments, generating actionable insights for smarter systems and services. These sensors are capable of capturing a broad array of physical and biological parameters, enabling applications in healthcare, smart cities, and industrial automation [66].

As IoE scales to billions of interconnected elements, it imposes stringent requirements on network performance, including scalability, reliability, and latency. Sixth-generation (6G) networks are positioned as a key enabler for IoE, offering ultra-reliable low-latency communication (URLLC) and enhanced mobile broadband (eMBB) to support real-time, high-throughput, and energy-efficient connectivity. Consequently, 6G will be essential in unlocking the full potential of IoE in future hyper-connected environments.

6.4. UAV based mobility

Unmanned Aerial Vehicles (UAVs), commonly known as drones, have been extensively utilized in defense and military operations for applications such as remotely operated aircraft and autonomous drones. Over the years, the versatility of UAVs has extended their use into civilian domains, including disaster relief, agricultural plantation protection, traffic monitoring, and environmental sensing. These applications have highlighted the potential of UAVs as indispensable tools in addressing critical societal and environmental challenges [67].

With the emergence of 6G and the Internet of Everything (IoE), UAVs are anticipated to evolve further into a UAV-to-Everything (U2X) network. This paradigm shift will enable UAVs to connect seamlessly with other devices and systems, expanding their utility beyond traditional communication. By leveraging advanced 6G capabilities such as ultra-low latency, high data rates, and massive connectivity, UAVs will optimize sensing applications and enhance their communication modes, unlocking their full potential for societal and industrial transformation [68].

6.5. Holographic telepresence

Holographic Telepresence (HT) is an emerging communication technology that enables the real-time projection of life-size, three-dimensional (3D) representations of people and objects, offering a sense of physical presence and immersion. By capturing, compressing, transmitting, and reconstructing visual information using high-speed